

FORECASTING & OPTIMIZATION

I turn uncertain demand into better planning, inventory and allocation decisions.

I build data-driven systems that connect prediction to the decision it is meant to improve — then report the result in cost, service level or allocation terms.

Strasbourg, France / ashfaquesalmantk@gmail.com / [LinkedIn](#) / [GitHub](#) / [CV \(PDF\)](#)

01 ABOUT

A **PhD-trained researcher** working in applied forecasting and operations research. My background is computational physics — large numerical systems, simulation, statistical analysis — and I bring that rigour to business problems. Most forecasting work stops at accuracy metrics; I care about the last mile, whether a forecast actually reduces stockouts, holding cost or idle capacity downstream.

02 WHAT I DO

01 Demand forecasting Demand forecasts with honest error estimates, built on walk-forward backtesting rather than random splits.	02 Inventory & planning Turning forecast error into safety stock, order-up-to levels and replenishment policy, scored on realised cost and fill rate.	03 Constrained optimisation Mixed-integer models for allocating scarce resources under budget, capacity and policy constraints, solved to proven optimality.
--	---	--

03 APPROACH

01 Frame the decision What decision are we actually trying to improve, and what does it cost today?	02 Baseline first Establish the cheapest model that works before adding complexity or buying a tool.	03 Validate honestly Walk-forward backtests, blind holdouts, and explicit failure modes. No random splits.	04 Report in business units Cost, service level, allocation, risk — plus what the model still cannot do.
---	--	--	--

04 WORK

Three representative systems

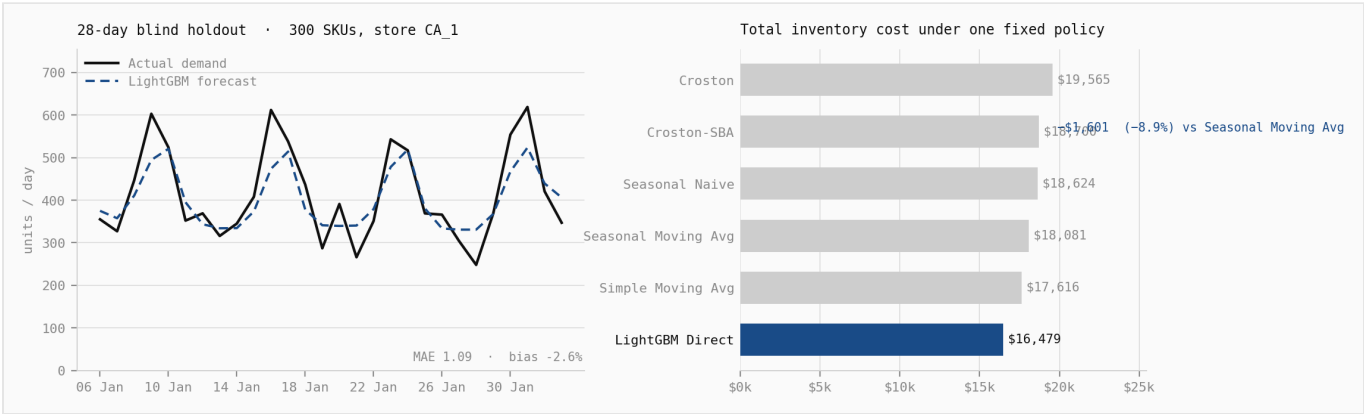
Retail demand forecasting → inventory decisions

CASE STUDY / FORECASTING + INVENTORY

Walmart M5 dataset · store CA_1 · 300 SKUs · 6 models · 28-day holdout · 2026

PROBLEM	A forecast is only worth having if it changes a decision downstream. Which demand model is accurate — and which one actually lowers inventory cost?
APPROACH	Causally-safe feature engineering (lags, rolling statistics, calendar, events, price) feeding recursive and direct LightGBM models, benchmarked against five operational baselines including seasonal naïve, moving averages and Croston. The selected forecast drives a periodic-review replenishment simulation: risk-period demand, error-based safety stock, order-up-to level, holding and stockout cost.
VALIDATION	Six walk-forward backtest windows for selection, then one blind 28-day holdout on unseen data. Every candidate is replayed through an identical replenishment policy, so any cost difference belongs to the forecast alone.
OUTCOME	The selected forecast was the cheapest of all six under that fixed policy — \$16,479 against \$18,081 for the seasonal moving average — and is served through PostgreSQL and a FastAPI endpoint rather than a notebook.

8.9%	- \$1,601	90.4%	1.09
LOWER TOTAL INVENTORY COST	VS. SEASONAL MOVING AVERAGE	FILL RATE, 28-DAY HOLDOUT	MAE UNITS / SKU / DAY



Blind holdout, store CA_1. Left: realised daily demand across 300 SKUs against the selected forecast. Right: every candidate replayed through one identical replenishment policy, so the cost spread is the forecast's doing — \$16,479 for LightGBM against \$18,081 for the seasonal moving average.

Optimising resource allocation under constraints

CASE STUDY / OPTIMISATION

Mixed-integer linear programming · SoFIFA 25 dataset · live Streamlit app · 2025

PROBLEM	Choosing a team under a wage budget, a fixed formation, positional hierarchies and locked-in selections is a constrained selection problem. Heuristic scoring gives a plausible answer, never a provably best one.		
APPROACH	Formulated as a MILP with role-aware binary variables $x[\text{player}, \text{role}]$, solved to optimality with PuLP. Sixteen roles are modelled, a player fills at most one, and budget, squad size, formation balance, age window and locked players all enter as hard constraints. A natural-language layer turns plain-text requirements into constraints.		
VALIDATION	Every solution is checked for constraint satisfaction and re-solved across scenario variations, so infeasible selections surface to the user rather than returning silently.		
OUTCOME	A working, publicly reachable application rather than a research script. The same pattern transfers directly to staffing, scheduling and capital allocation.		
Optimal	16	6	Live
SOLUTION STATUS, NOT HEURISTIC	ROLES MODELLED PER PLAYER	CONSTRAINT FAMILIES ENFORCED	DEPLOYED STREAMLIT APPLICATION

Production ML under uncertainty

CASE STUDY / PRODUCTION ML

800k records · severe class imbalance · FastAPI + Docker · 2025

PROBLEM	A content-classification service facing a <10% minority class and noisy, out-of-distribution traffic. A single hard threshold either floods the review queue or silently drops real violations.		
APPROACH	TF-IDF + linear SVM with Platt scaling for probability calibration, wrapped in a three-tier confidence router: high-confidence predictions are auto-handled, low-confidence ones escalate to review, the ambiguous middle band is logged for retraining. Containerised as a FastAPI service with structured request logging.		
VALIDATION	Recall deliberately optimised over precision, because the cost of a missed violation outweighs a false positive. Load behaviour stress-tested separately from model accuracy.		
OUTCOME	A service that degrades predictably under distribution shift instead of failing silently — and generates its own training signal from the escalation queue.		
800k	85%	61%	3
RECORDS PROCESSED	RECALL ON MINORITY CLASS	PRECISION	CONFIDENCE ROUTING TIERS

05 EXPERIENCE

Scientific Researcher

Oct 2022 – Oct 2025

IPCMS, University of Strasbourg · CNRS

- Ran large-scale finite-element micromagnetic simulations of topologically complex structures, analysing **10M+ data points** across high-dimensional 3D meshes.
- Built automated Python pipelines for extraction, transformation and analysis of simulation output, **cutting analysis time by roughly 60%** across repeated runs.

06 EDUCATION & RESEARCH

PhD, Condensed Matter Physics

2022 – 2025

IPCMS & University of Strasbourg, France

- Dissertation on spin-wave dynamics in topologically complex magnetic structures. Trained in large-scale computation, algorithm optimisation and research-grade data pipelines.

MSc, Physics

2018 – 2020

Cochin University of Science & Technology, India · statistical mechanics, computational physics, numerical methods

BSc, Physics

2015 – 2018

University of Calicut, India

07 WRITING

- **Rotating spin wave modes in nanoscale Möbius strips.** *NPJ Spintronics*, 4(4), 2026. doi.org/10.1038/s44306-025-00118-8
- Technical writing on forecasting, optimisation and the research-to-industry transition — medium.com/@cmtwskb

08 TOOLKIT

LANGUAGES	Python, SQL, LaTeX · English (fluent), French (learning)
FORECASTING	Walk-forward backtesting, ARIMA, SARIMAX, exponential smoothing, Prophet, LightGBM, WRMSSE
OPTIMISATION	MILP (PuLP), constraint programming, OR-Tools, operations research
ML, STATS & SCIENTIFIC	scikit-learn, XGBoost, probability calibration, imbalanced classification, PCA, NumPy, SciPy, pandas, FFT, spatial interpolation
DELIVERY	FastAPI, Docker, PostgreSQL, Streamlit, Git, Linux, Power BI

CONNECT

I take on a few engagements at a time — usually a forecasting or optimisation problem where the data already exists and the impact is measurable. A first conversation is usually enough to tell whether I can help.

ashfaquesalmantk@gmail.com [LinkedIn](#) [GitHub](#)